A Stabilization Counterexample to the General Form of Problem 4.8 in the K3 Problem List

Abstract

Problem 4.8 in *K3: A New Problem List in Low-Dimensional Topology* asks whether Fintushel–Stern knot surgery on a fixed pair (X, T) distinguishes prime knots up to mirroring. As the problem is written, the ambient manifold is not required to be irreducible or minimal. We show that this general formulation has a direct counterexample. Take a regular elliptic fiber $F \subset E(2)$, set

$$X = E(2) \# (S^2 \times S^2),$$

and choose the prime torus knots $K_1 = T(2,3)$ and $K_2 = T(2,5)$. Knot surgery is local with respect to the connected sum away from F, while Baykur's one-stabilization theorem gives

$$E(2)_K \# (S^2 \times S^2) \cong E(2) \# (S^2 \times S^2)$$

for every knot K. Consequently $X_{K_1} \cong X_{K_2}$ although the two knots are neither equivalent nor mirrors. This refutes the literal general statement. It does not settle the separate case $X = E(2)$.

1 The problem and the counterexample

The general form of Problem 4.8 fixes a closed simply connected smooth 4–manifold X and a smoothly embedded torus $T \subset X$ satisfying

$$\pi_1(X - T) = 1, \qquad [T]^2 = 0.$$

For a knot $K \subset S^3$, let X_K denote Fintushel–Stern knot surgery along T. The problem asks whether, for prime knots K_1, K_2, a diffeomorphism $X_{K_1} \cong X_{K_2}$ forces K_1 to equal K_2 or its mirror.

Theorem 1. *The answer to the general form of Problem 4.8 is no. More precisely, let*

$$Y = E(2), \qquad Z = S^2 \times S^2, \qquad X = Y \# Z,$$

and let $F \subset Y \subset X$ be a regular elliptic fiber, with the connected sum performed away from F. For

$$K_1 = T(2,3), \qquad K_2 = T(2,5),$$

the pair (X, F) satisfies all hypotheses of the problem, K_1 and K_2 are prime and are not equivalent up to mirroring, but

$$X_{K_1} \cong X_{K_2}.$$

The proof is divided into the ambient-manifold check, the knot check, the locality of knot surgery, and the stabilization argument.

2 The ambient pair

Lemma 2. *The manifold $X = E(2)\#(S^2 \times S^2)$ is closed, simply connected, and smooth. The regular elliptic fiber $F \subset X$ is a square-zero torus and*

$$\pi_1\big(X - \operatorname{int}\nu(F)\big) = 1.$$

Proof. The elliptic surface $E(2)$ is the simply connected K3 surface. A regular elliptic fiber has a product tubular neighborhood $F \times D^2$, and hence $F^2 = 0$. The standard elliptic fibration has a section. After deleting a regular fiber, the base becomes a disk; the section removes the base-loop generator, while vanishing cycles parallel to the two standard generators a, b of the torus fiber normally kill both fiber generators. Thus

$$\pi_1\big(E(2) - \operatorname{int}\nu(F)\big) \cong \langle a, b \mid [a, b] = 1\rangle / \langle\langle a, b\rangle\rangle = 1.$$

Perform the connected sum with $Z = S^2 \times S^2$ in a ball disjoint from F. Since Z is simply connected, van Kampen's theorem gives both

$$\pi_1(X) = 1 \quad \text{and} \quad \pi_1\big(X - \operatorname{int}\nu(F)\big) = 1.$$

The connected sum does not alter the embedding or normal framing of F. $\square$

3 The two knots

Lemma 3. *The knots $K_1 = T(2, 3)$ and $K_2 = T(2, 5)$ are prime, and K_1 is equivalent to neither K_2 nor the mirror of K_2.*

Proof. Schultens' bridge-number theorem gives

$$b\big(T(p, q)\big) = \min\{|p|, |q|\},$$

so $b(K_1) = b(K_2) = 2$. If either knot were a connected sum of two nontrivial knots, bridge-number additivity would give

$$b(J\#L) = b(J) + b(L) - 1 \geq 3,$$

a contradiction. Hence both knots are prime.

Their symmetrically normalized Alexander polynomials are

$$\Delta_{K_1}(t) = t - 1 + t^{-1}, \qquad \Delta_{K_2}(t) = t^2 - t + 1 - t^{-1} + t^{-2}.$$

The respective breadths are 2 and 4. Alexander breadth is preserved by knot equivalence and by mirroring, since $\Delta_{\overline{K}}(t) = \Delta_K(t^{-1})$ up to multiplication by a unit. Therefore K_1 is neither K_2 nor its mirror. $\square$

4 Locality of knot surgery

Write $E_K = S^3 - \operatorname{int}\nu(K)$. Knot surgery on Y along F is the cut-and-paste manifold

$$Y_K = \big(Y - \operatorname{int}\nu(F)\big) \cup_{\varphi_K} (S^1 \times E_K),$$

with the standard Fintushel–Stern zero-surgery boundary identification.

Lemma 4. *For every knot K, there is a canonical diffeomorphism*

$$\lambda_K : X_K \longrightarrow Y_K \# Z.$$

Proof. Choose the connected-sum balls defining $X = Y \# Z$ in $Y - \operatorname{int} \nu(F)$ and in Z. The knot-surgery replacement of $\nu(F)$ is disjoint from the connected-sum neck. Regrouping the same glued pieces gives

$$\begin{aligned} X_K &= \big(((Y - \operatorname{int} B) - \operatorname{int} \nu(F)) \cup_{\varphi_K} (S^1 \times E_K)\big) \cup_{\partial B} (Z - \operatorname{int} B') \\ &\cong \big((Y - \operatorname{int} \nu(F)) \cup_{\varphi_K} (S^1 \times E_K)\big) \# Z \\ &= Y_K \# Z. \end{aligned}$$

The diffeomorphism is the identity on each displayed piece, together with the standard associativity diffeomorphism on collars of the gluing boundaries. No comparison of invariants is used. □

5 Baykur stabilization and conclusion

We use the following published theorem.

Theorem 5 (Baykur's one-stabilization theorem)**.** *Let M be a compact simply connected smooth 4–manifold and let $T \subset M$ be a square-zero torus with simply connected complement. For every knot $K \subset S^3$,*

$$M_K \# (S^2 \times S^2) \cong M \# (S^2 \times S^2).$$

The first conclusion of this theorem has no non-spin hypothesis; that hypothesis occurs only in Baykur's alternative $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ stabilization statement.

Proof of Theorem 1. Lemma 2 shows that (Y, F) satisfies every hypothesis of Theorem 5. Hence for $i = 1, 2$ there is a diffeomorphism

$$\beta_i : Y_{K_i} \# Z \longrightarrow Y \# Z = X.$$

By Lemma 4, there is also a diffeomorphism

$$\lambda_i : X_{K_i} \longrightarrow Y_{K_i} \# Z.$$

Consequently

$$\Phi = \lambda_2^{-1} \circ \beta_2^{-1} \circ \beta_1 \circ \lambda_1 : X_{K_1} \longrightarrow X_{K_2}$$

is a diffeomorphism. Lemmas 2 and 3 verify all remaining hypotheses and show that K_1 and K_2 are not equivalent up to mirroring. Thus (X, F, K_1, K_2) is a counterexample to the general form of Problem 4.8. □

6 Scope and mathematical content

The construction uses the reducible ambient manifold $E(2) \# (S^2 \times S^2)$. This is permitted by the literal general statement, which imposes no irreducibility, minimality, or indecomposability condition on X. The counterexample therefore settles that general formulation negatively.

The separate question in the remarks asking specifically for $X = E(2)$ is not settled by this construction. For the K3 surface itself, the stabilizing $S^2 \times S^2$ summand is unavailable, so Baykur's theorem does not identify $E(2)_{K_1}$ with $E(2)_{K_2}$.

The proof introduces no new stabilization theorem. Its mathematical content is the observation that Baykur's published theorem, combined with the full generality of the quantifier in Problem 4.8 and the locality of knot surgery, produces an explicit fixed-pair counterexample.

7 Machine-verification status

The accompanying finite certificate is replayed by `verify_certificate.py`. It checks the structured witness, the two Alexander polynomials, the domains and codomains of the displayed diffeomorphism composition, the source transcriptions used in the theorem applications, and the typed Jacobian polynomial-factor output. The campaign math ledger and proof graph also pass their structural check, with no open dependency below the goal.

A companion Lean 4 development, `Problem48.lean`, formalizes the exact logical composition used in the final step. It takes as explicit hypotheses the ambient admissibility statement, the locality diffeomorphisms, and Baykur's stabilization diffeomorphisms. From those hypotheses it constructs the named prime-knot witness and proves that the two knot-surgery manifolds are diffeomorphic by symmetry and transitivity. The theorem is paired with `statement_fidelity.md`.

Argus's native `lean_evidence` pipeline compiled this source with Lean 4.31.0, found no proof holes, completed its environment axiom audit with exit code zero, and recorded the formal claim as `closed_kernel`. Lean's own axiom report states that the formal theorem depends on no axioms.

This is a kernel check of the complete implication from the three source-backed topological inputs to the counterexample witness. It is not a from-first-principles formalization of smooth 4–manifold topology: Baykur's theorem, the elliptic-fiber complement calculation, and the locality lemma occur transparently as hypotheses of the Lean theorem and remain discharged by the cited mathematical sources and independent human review.

References

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